Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI-based Telematics Terminals and Integrated Vehicle Management Platform for Intelligent Mobility with Hybrid Connectivity and Carbon Reduction**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	ITS / Telematics	*Telematics Terminals and Devices*	*ITS / Telematics*	*Telematics Application Services*
Concept of Proposed Technology	**Core Concept** This technology is an AI-based telematics terminals and devices solution that collects, transmits, and analyzes location, driving status, and environmental data from vehicles and mobile assets in real time. **Key Functions and Potential Applications** GNSS positioning data, vehicle status monitoring, and environmental sensor data are integrated with LTE-based terrestrial communication, while a hybrid communication architecture incorporating satellite communication (Starlink) is applied in areas where terrestrial networks are unavailable, enabling the system to achieve a data transmission reliability of over 95% regardless of communication conditions.			

	Based on the collected data, the project implements an integrated telematics platform that supports vehicle tracking, fleet and logistics management, mobility service optimization, and predictive maintenance. In addition, AI-based analytics are used to analyze driving patterns, idling, and congestion segments, thereby enabling intelligent mobility services that optimize fuel consumption and reduce carbon emissions.
Rationale for Joint R&D Project	This research and development project is being pursued to secure the global applicability and technological reliability of AI-based telematics terminals and integrated vehicle management platforms through international joint R&D with a university located in the United States. The United States, with its vast territory and diverse road, climate, and traffic conditions, provides an optimal testbed for validating and advancing telematics technologies. In particular, experience in collecting and analyzing vehicle data in mixed environments—urban, suburban, rural, and communication shadow areas—is essential. By applying a hybrid communication architecture utilizing satellite communication (Starlink) in areas with limited terrestrial connectivity, the project aims to verify technologies that ensure a data transmission reliability of over 95% across diverse vehicle operating environments in the United States. Furthermore, leveraging the U.S. university's expertise in vehicle operation data analysis and AI algorithm research, the project seeks to improve the accuracy and generalization performance of AI analytics models based on real-world U.S. vehicle data. Through this approach, the project aims to develop AI-based carbon reduction services that optimize fuel consumption and reduce carbon emissions, while securing telematics technologies that can be readily applied to global mobility and logistics environments. **Why Global Collaboration?** Purdue University, a leading overseas research institution, is a manufacturing-focused engineering university located in Indiana, a region surrounded by numerous manufacturing companies, and serves as an industrial hub in the United States. The university has already been collecting meaningful equipment data for smart manufacturing and future mobility and digital transformation in the surrounding area and utilizing them for AI-based research. Purdue University is a globally recognized research institution with strong foundational technologies in mobility, supported by specialized centers such as CCAT, TASI, and PURT, enabling advanced research and industry collaboration. (1) Purdue maintains strong collaborative relationships with medium- to large-scale enterprises equipped with factories, facilities, and production lines; (2) it operates specialized research centers capable of conducting testbed-based research on

	manufacturing, mobility, and physical AI; (3) it is strategically located in a major transportation hub in the U.S., making it the only academic institution where Korean companies can directly identify technological trends and market demands in the local environment. Therefore, global collaboration is essential.
Key Words	**1) Telematics terminals and devices** **2) Hybrid communications (terrestrial-satellite integration, Starlink)** **3) High-reliability data transmission (95%+ transmission rate)** **4) AI-based vehicle data analysis** **5) Carbon-reduction mobility services**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	**R&D Objectives** The ultimate objective of this R&D project is to develop AI-based telematics terminals and an integrated vehicle management platform that can operate reliably across diverse global transportation environments, and to realize intelligent mobility services that enable high-reliability data transmission and carbon reduction through hybrid communication and AI analytics technologies. To achieve this objective, the project aims to verify and further advance the developed technologies through international joint R&D with a U.S.-based university, utilizing real-world vehicle operation data collected in the United States, and to secure globally applicable telematics technologies that can be validated through overseas pilot demonstrations. **Detailed R&D Objectives** **1) Development of Core Technologies for AI-based Telematics Terminals and Devices** • Development of telematics terminal prototypes capable of real-time collection of location, driving status, and environmental data from vehicles and mobile assets • Design of low-power, high-reliability hardware architectures considering vehicle operating environments **2) Development of Hybrid Terrestrial–Satellite Communication Technology** • Design of a hybrid communication architecture integrating terrestrial mobile networks (e.g., LTE) and satellite communication (Starlink) • Implementation of data transmission and recovery mechanisms that minimize data loss and ensure a data transmission reliability of over 95%, even in communication shadow areas			

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	3) Acquisition of Data Collection and Analysis Technologies for Global Vehicle Environments • Collection and utilization of vehicle operation data under diverse road, traffic, and climate conditions in the United States • Enhancement of accuracy and generalization performance of analytics models through collaboration with a U.S.-based university, using real-world vehicle operation data **4) Development of AI-based Vehicle Data Analytics and Predictive Services** • Development of AI algorithms for analyzing driving patterns, idling behavior, congestion segments, and operational efficiency • Implementation of intelligent analytics services that support vehicle condition anomaly detection and predictive maintenance **5) Implementation of AI-driven Carbon Reduction Mobility Services** • Development of carbon reduction logic through optimization of fuel consumption and reduction of unnecessary driving based on vehicle operation data • Proposal of mobility and logistics service models that improve operational efficiency and support ESG compliance using AI analytics results **6) Verification of Global Applicability through Pilot Demonstration** • Verification of integrated performance of terminals, communication systems, and platforms in pilot-scale environments beyond laboratory-level testing • Establishment of a foundation for global market application and commercialization through overseas pilot demonstrations (PoC)
R&D Target Specifications	**1) Hybrid Communication–Based Data Transmission Reliability** • **Target Description:** Implement a hybrid communication architecture integrating terrestrial mobile communication (e.g., LTE) and satellite communication (Starlink) to ensure stable data transmission regardless of communication conditions. • **Target Performance:** Achieve a data transmission success rate of **over 95%** relative to the total collected data. **2) Telematics Terminal Data Collection and Positioning Accuracy** • **Target Description:** Develop telematics terminals capable of real-time collection of location, driving status, and environmental data from vehicles and mobile assets. • **Target Performance:** Achieve **GNSS-based positioning accuracy within ±5 meters**.

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	3) AI-based Vehicle Data Analytics Accuracy • **Target Description:** Develop AI algorithms for analyzing vehicle operation data (e.g., driving patterns, idling behavior, congestion segments), and validate analytics performance using pilot data including real-world vehicle data collected in the United States. • **Target Performance:** Achieve **AI analytics accuracy of 90% or higher**. **4) Implementation of AI-based Carbon Reduction Services** • **Target Description:** Identify fuel inefficiencies, unnecessary driving, and idling through AI analytics, and implement mobility services capable of reducing carbon emissions based on these insights. • **Target Performance:** Derive and quantify **carbon reduction effects through improved vehicle operation efficiency**. **5) Integrated Platform Demonstration and Technology Maturity Advancement** • **Target Description:** Conduct pilot-scale demonstrations by integrating telematics terminals, hybrid communication, AI analytics, and the vehicle management platform. • **Target Performance:** Achieve **Technology Readiness Level (TRL) advancement from TRL 4 to TRL 6**, and verify applicability through **overseas pilot demonstrations (PoC)**.
R&D Plan	This R&D project aims to secure the global applicability of AI-based telematics terminals and an integrated vehicle management platform. To this end, the project will be carried out in a phased manner, focusing on the implementation of hybrid communication, high-reliability data transmission, AI analytics, and carbon reduction services. The R&D activities are structured into three phases**:** **(1) core technology development → (2) system integration → (3) pilot demonstration.** **Phase 1: Core Technology Design and Development** ***(Achieving TRL 4)*** • Design of the hardware architecture for telematics terminals • Design of a hybrid communication architecture integrating terrestrial mobile communication (e.g., LTE) and satellite communication (Starlink) • Design of baseline AI-based data analytics models **Phase 2: System Integration and Prototype Development** ***(Achieving TRL 5)*** • Development of telematics terminal prototypes • Implementation and performance validation of hybrid communication • Development of an integrated vehicle management platform • Enhancement and optimization of AI analytics algorithms **Phase 3: Pilot Demonstration and Validation**

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	(Achieving TRL 6) • Deployment and demonstration in pilot-scale environments • Verification of key performance indicators • Validation of AI-based carbon reduction effects • Preparation for technology advancement and linkage to commercialization

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	As the technical lead of this international joint R&D project, Purdue University serves as the overall technical coordinator, leading the development of core technologies and system integration for AI-based telematics terminals and an integrated vehicle management platform. The institute plans and manages the entire R&D process with a focus on the implementation of hybrid communication, high-reliability data transmission, AI analytics, and carbon reduction services, while overseeing technical collaboration among domestic and international participating organizations. **Detailed Roles and Responsibilities** **1) Overall System Architecture Design and Technical Leadership** • Design of the overall system architecture encompassing telematics terminals, communication systems, AI analytics, and the vehicle management platform • Establishment of the R&D roadmap and formulation of technology integration strategies **2) Development of Core Technologies for AI-based Telematics Terminals** • Design and prototype development of telematics terminal hardware for vehicles and mobile assets • Development of data acquisition technologies for GNSS, vehicle status, and environmental sensors • Implementation of low-power, high-reliability terminal architectures considering vehicle operating environments **3) Design and Implementation of Hybrid Communication Technologies** • Design of a hybrid communication architecture integrating terrestrial mobile communication (e.g., LTE) and satellite communication (Starlink) • Development of transmission and recovery mechanisms that minimize data loss and ensure a data transmission reliability of over 95%, even in communication shadow areas **4) Development of AI-based Data Analytics and Carbon Reduction Services** • Development of AI analytics algorithms based on vehicle operation data • Development of predictive maintenance and operational efficiency improvement technologies through analysis of driving patterns, idling behavior, and congestion segments • Implementation of carbon emission reduction indicators and service models utilizing AI analytics results **5) System Integration, Performance Validation, and Technology Advancement** • Verification of integration and interoperability among terminals, servers, and the platform • Performance evaluation in pilot-scale environments and validation of technologies at TRL 6 • Execution of technology improvement and advancement based on pilot demonstration results
Expected R&R of Korean SME Partner	As a key implementation entity in the international joint R&D project, the Korean SME partner is responsible for the productization and real-world deployment of AI-based telematics terminals and devices. Based on the core technologies developed by the research institute, the partner carries out prototype manufacturing, field validation, and preparation for mass production and commercialization.

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	Detailed Roles and Responsibilities **1) Prototype Manufacturing and Product Design of Telematics Terminals** • Manufacture of telematics terminal prototypes based on the technical specifications designed by the research institute • Execution of product-oriented design, including mechanical design, PCB configuration, and power architecture • Ensuring durability and operational stability considering vehicle installation and operating environments **2) Communication Module Integration and Terminal Interface Support** • Integration and support of terrestrial mobile communication modules (e.g., LTE) into telematics terminals • Support for terminal–communication interface integration to implement hybrid communication architectures • Verification of stable terminal operation under changing communication conditions **3) Field Testing and Pilot Demonstration Support** • Conducting application tests of telematics terminals in real vehicles or equivalent operational environments • Verification of data collection and transmission stability, as well as operational usability • Providing feedback on demonstration results and improvement items to the research institute **4) Review of Mass Production Feasibility and Certification/Regulatory Compliance** • Review of manufacturing processes and analysis of cost structures considering mass production of terminals • Support for compliance with domestic and international certifications, communication regulations, and vehicle-related regulatory requirements • Incorporation of product requirements for entry into global markets **5) Commercialization and Market Linkage Activities** • Analysis of market requirements in mobility, logistics, and vehicle management sectors • Specification and development of business models based on R&D outcomes • Collaboration with overseas partners for joint pilot demonstrations and PoC-based commercialization

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1) Technological Outcomes** The project secures core technologies for AI-based telematics terminals and an integrated vehicle management platform, and implements high-reliability data transmission technologies based on hybrid communication that integrates terrestrial mobile networks and satellite communication (Starlink). • Securing telematics terminal technologies capable of achieving **data transmission reliability of over 95%**, regardless of communication conditions

- Implementation of integrated telematics technologies that enable real-time collection and analysis of location, driving status, and environmental data from vehicles and mobile assets
- Securing globally applicable data analytics technologies through validation of AI models using real-world vehicle operation data from the United States
- Achieving system-level technological completeness by integrating terminals, communication systems, and platforms (**TRL 6 level**)

2) Commercial and Economic Impact

The telematics terminals and platform represent technologies that can be immediately commercialized in mobility, logistics, vehicle management, and transportation service sectors, contributing to global market expansion and new business creation for Korean SMEs.

- Creation of new revenue streams in vehicle management, logistics management, and mobility service sectors
- Expansion of overseas pilot demonstrations (PoC) and exports of hybrid communication–based telematics terminals
- Expansion of platform-based subscription and service-oriented business models
- Establishment of a foundation for global business collaboration through joint demonstrations with U.S. universities and overseas partners

3) Industrial Impact

The project contributes to technological advancement and enhanced competitiveness of the intelligent transportation and advanced mobility industries.

- Dissemination of telematics technologies capable of operating in global environments, including communication shadow areas
- Improvement of overall industrial productivity through data-driven optimization of vehicle and logistics operations
- Expansion of international joint R&D and technology collaboration models centered on SMEs
- Strengthening of global standard compliance capabilities of domestic telematics and mobility technologies

4) Environmental and Social Impact

The project contributes to the creation of environmental and social value through the implementation of AI-based data analytics–driven carbon reduction mobility services.

- Optimization of fuel consumption and reduction of carbon emissions through analysis of driving patterns, idling behavior, and inefficient driving
- Expansion of environmentally friendly transportation services through improved efficiency of vehicle and logistics operations
- Presentation of mobility and logistics service models aligned with ESG requirements
- Reduction of social costs through enhanced traffic safety and operational reliability

6. References

(1) Guo, Jingwei, et al. "Spatiotemporal analysis of CO_2 emissions from heavy-duty trucks in belt and road port transportation using GPS trajectory data." *Resources, Conservation and Recycling* 223 (2025): 108540.

(2) Chong, Yanyi, et al. "Research Progress on CO2 Emission Simulation for Heavy-Duty Commercial Vehicles." *Sustainability* 17.7 (2025): 2909.

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(3) IEA 의 '2023 년 이산화탄소 배출량(CO2 Emissions in 2023)' 보고서

(4) 김판수, 유준규. "위성-지상 직접통신 기술 동향." *전자파기술* 35.2 (2024): 35-45.